Morphological Resonance Synthesis (MRS): Language Construction Without Tokens, Gradients, or Learned Embeddings

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The byte-lane sensory layer is currently input-only: bytes enter the substrate through learned couplings, but output leaves it as retrieved label strings. That retrieval is the last token-shaped operation in the pipeline. MRS removes it by making the byte layer bidirectional and giving it the same three phase mechanisms the concept layer already has — chirality on input, phase-consistent consolidation, and phase-collapse projection on output. Word forms are then synthesised *by sweeping a byte-phase field rather than* retrieved *from a table.*

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0. The Remaining Asymmetry

The CMD substrate ingests raw bytes. There is no tokeniser: each of 256 `ByteLane` oscillators is driven by its byte value, and lanes reach the concept field only through Hebbian couplings $w[b][c]$ that begin at zero. On the input side, the architecture is already sub-lexical and token-free.

The output side is not. `readout_phase_collapse()` selects concept cavities by phase and then prints `si.graph.nodes[label_key].label` — a **stored string**. The physics decides *which* concepts

speak and *in what order*, but the *form of each word* is a lookup.

This asymmetry has three consequences:

1. **No morphology.** The engine can emit `walk` if `walk` is a stored label, and `walked` if `walked` is separately stored, but it cannot construct `walked` from `walk`. There is no mechanism by which stem and inflection compose.
2. **No novel forms.** Any word form absent from the label table is unreachable, regardless of how strongly the substrate's state implies it.
3. **A token-shaped residue.** The label table is a finite inventory of atomic surface strings indexed by graph key. Functionally that is a vocabulary, and its presence is what makes the pipeline token-like at the boundary even though the interior is not.

MRS closes the asymmetry. The mechanism is a direct reuse — one layer down — of machinery the substrate already contains.

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1. Architecture

Let the byte-lane population be L_b for $b \in [0, 256)$, each an oscillator with displacement x_b, velocity v_b, and natural frequency $\omega_b = 3.0 + b/256$ (as in the existing CMD sensory layer). Let $w[b][c]$ be the learned byte→concept binding weights.

MRS adds four mechanisms.

1.1 Reverse Drive (concept → lane)

The existing forward path applies a force from lane to concept:

$$F_c \mathrel{+}= \frac{w[b][c] \cdot x_b}{m_c}$$

MRS adds the transpose, using the **same weights**:

$$F_b \mathrel{+}= \eta_{\text{rev}} \cdot w[b][c] \cdot x_c$$

with η_{rev} a reverse-gain scalar (default 0.4). No new storage: the coupling matrix is used in both directions, making the byte↔concept layer a bidirectional associative memory implemented as coupled oscillators rather than as a matrix multiplication.

Consequence: when concept cavities resonate, the byte lanes whose bindings to those concepts are strongest resonate in turn. The lanes that light up are the letters of the words those concepts were historically written with.

1.2 Lane Chirality

Byte lanes acquire the CPME §1.1 arrival timestamp:

$$t_{\text{arrival},b} \leftarrow t_{\text{sim}} \quad \text{on each drive of } L_b$$

During input of `hi`, lane $L_{\texttt{h}}$ is driven before $L_{\texttt{i}}$ and therefore accumulates more phase. Letter order within a word becomes a phase gradient across the lane population — exactly the mechanism CPME already uses for word order across concepts, applied one level down.

1.3 Lane-to-Lane Gauge Tunnels

A sparse set of lane↔lane tunnels is created on co-occurrence, each carrying a CPSB-style gauge offset $\Psi_{bb'}$ and consolidated by the CPME-II phase-consistent rule:

$$\Delta w_{bb'} \;\propto\; v_b \cdot v_{b'} \cdot \cos\!\big(\theta_b - \theta_{b'} - \Psi_{bb'}\big)$$

where $\theta_b = \operatorname{atan2}(v_b, \omega_0 x_b)$. Unlike the concept layer, $\Psi_{bb'}$ is **not** assigned from a relation table — it is *learned* as the running circular mean of observed phase differences:

$$\Psi_{bb'} \leftarrow \arg\!\Big(\lambda\, e^{i\Psi_{bb'}} + (1-\lambda)\, e^{i(\theta_b - \theta_{b'})}\Big)$$

This is the substrate's representation of *sequence*: the tunnel between `h` and `i` stores the typical phase lead of `h` over `i`. Nothing about letters, words, or grammar is encoded declaratively; only phase offsets between oscillators.

1.4 Byte-Level Phase-Collapse Projection

Output is generated by the CPME §1.6 sweep applied to the lane population. Carrier phase ϕ_{carrier} advances monotonically across $[0, 2\pi)$ in K_{byte} steps; at each step the emitted byte is the arg-max of

$$w_b(\phi_{\text{carrier}}) \;=\; |x_b| \cdot \mathrm{ReLU}(P_b) \cdot \exp\!\big(-\kappa_b(\phi_{\text{carrier}} - \theta_b)^2\big)$$

with P_b the lane's envelope momentum as in CPME §1.2.

Critical divergence from the concept sweep. In the concept layer, a winner is suppressed from later sweep steps — a concept appears once in a clause. In the byte layer, **winners are not suppressed**, and the sweep may make multiple passes. Letters repeat within words (`banana`, `letter`, `success`); a suppression rule would make repeated graphemes unreachable. The difference is physically motivated by what each layer represents, not introduced as an exception.

Word boundaries emerge from amplitude gaps: when the swept weight falls below the emission floor for a contiguous span of carrier phase, a boundary is emitted. Boundaries are therefore not delimiters in a grammar but silences in a resonance field.

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2. Why This Is Not Tokens, ML, DL, or an LLM

Not tokenisation. There is no vocabulary, no merge table, no subword inventory, no `<unk>`. Input is 256 byte lanes; output is a swept byte sequence. The only inventory is the byte alphabet itself, which is a property of the encoding, not a learned artefact.

Not machine learning in the gradient sense. There is no loss function, no gradient, no backpropagation, no optimiser, no train/test split. All adaptation is the local Hebbian rule of §1.3, which uses only quantities available at the two endpoints of a single tunnel at a single tick. Nothing computes a derivative of a global objective.

Not deep learning. There is no layer stack in the DL sense — no weight matrices applied in sequence, no nonlinearity between layers, no depth over which credit is assigned. The byte layer and concept layer are two populations of coupled oscillators exchanging forces bidirectionally through one shared coupling matrix.

Not a language model. No distribution over next symbols is ever formed. There is no softmax, no sampling temperature, no autoregressive conditioning on prior output. The emitted sequence

is the projection of a static field state onto a swept phase axis — a geometric readout, not a sequential prediction. Given the same field state the same sequence is produced deterministically, and the sequence is generated *simultaneously* in the sense that all lanes are already in their phases before the sweep begins.

What it is: a bidirectional associative resonance memory between a 256-oscillator sub-lexical layer and an N-oscillator conceptual layer, read out by phase-swept projection.

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3. Where the Storage Lives (Translation Capacity Lemma)

The Translation Capacity Lemma (SI §3.1) states that any system discriminating N outcomes must store $\log_2 N$ bits somewhere. MRS does not evade this and does not claim to. The storage relocates:

Before MRS	After MRS
N label strings in `graph.nodes[].label`	$256 \times N$ byte→concept couplings $w[b][c]$
—	$\leq 256^2$ lane↔lane couplings and offsets $\Psi_{bb'}$

The relocation is the substantive change. A label table stores each surface form *atomically and independently*: `walk` and `walked` share no representation, so knowing one contributes nothing to producing the other. A coupling matrix stores forms *distributedly and compositionally*: the lanes and phase offsets that produce `walk` are largely the same ones that produce `walked`, differing by the additional offsets contributed by the co-active inflectional concept.

Generalisation to unseen forms therefore becomes possible in principle — not because information was created, but because the information was stored in a factored representation instead of an atomic one. The bit count is not reduced; its *organisation* is what changes.

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4. Expected Dynamics

Prompt: `metformin lowers glucose`

1. **Byte injection.** Lanes for each character are driven in arrival order; t_{arrival} records the chirality gradient. Lane↔lane tunnels consolidate the observed offsets (`m` leads `e` leads `t`…).
2. **Concept resonance.** Forward couplings $w[b][c]$ excite the `metformin`, `lowers`, `glucose` cavities; propagation through the ontology raises downstream cavities (`ampk`, `glucose_uptake`).
3. **Acoustic-mass muting.** Directly-driven concepts are muted (CPME §1.3); downstream concepts are not.
4. **Reverse drive.** The surviving downstream concepts drive byte lanes through $w[b][c]^\top$. Lanes corresponding to the letters of `ampk`, `glucose`, `uptake` resonate — *without those strings being retrieved.*
5. **Byte sweep.** The carrier phase sweeps the lane population. Lanes emit in phase order, with boundaries at amplitude gaps. Output is a byte sequence assembled from resonance.

Morphological composition. If the inflectional concept for past-tense has been bound to the lanes `e`,`d` through prior exposure, and it is co-active with `lower`, the reverse drive raises both the stem lanes and the suffix lanes. The sweep — ordered by learned phase offsets — emits `lowered`, a form that need never have been stored as a label.

4b. Empirical Result: Single-Byte Lanes Are Insufficient (measured)

The single-byte formulation of §1.1–§1.4 was implemented and tested. **It does not work, and the reason is representational rather than a matter of tuning.**

Recognition was attempted by grounding every concept label to the 256 lanes at load time (binding weight = inverse byte document-frequency across the lexicon) and then recognising a query by driving its bytes and reading which concepts accumulated energy. Measured on a 2,240 -concept graph:

query	recognised as	correct
`metfromin`	`from`, `front`, `rfe`	×
`metformin`	`from`, `front`, `rfe`	×
`insluin`	`uni`, `ull`, `sil`	×
`fissoin`	`fission`	✓

Two distinct causes, and separating them matters:

(a) A normalisation error. Binding weight was divided by $\sqrt{|\text{label}|}$. For the query `metformin`, the substring `from` scored $S/2$ while `metformin` itself scored $S/3$ — the extra bytes $\{e, t, i, n\}$ are common and contribute ≈ 0 weight, but the length penalty applies regardless. A substring therefore outscored the whole word, which is why recognition failed even on *correctly spelled* input. This was a genuine bug.

(b) A representational limit. Removing the length division and replacing it with symmetric coverage (harmonic mean of query-recall and label-precision over byte sets) fixed the substring case and broke a different one: long labels sharing many bytes with the query (`intent_from_marker_label`) began to win, because a concept bound to more lanes accumulates more raw energy regardless of coverage. Each subsequent scoring adjustment relocated the failure rather than removing it.

That pattern — a defect that moves under every reweighting but never disappears — is the signature of insufficient representation. The decisive case is `metformin` versus `metfromin`: **they have identical byte sets.** A bag of distinct bytes cannot distinguish a word from its own transposition, so no weighting function over that bag can succeed. This is §5.3 (homograph collision) manifesting not as an edge case but as the dominant failure.

4b.1 Required correction: bigram lanes

The sensory layer must carry ordering information. Replacing the 256 single-byte lanes with 256^2 ordered-pair lanes supplies it:

$$\texttt{metformin} \rightarrow \{\texttt{me}, \texttt{et}, \texttt{tf}, \texttt{fo}, \texttt{or}, \texttt{rm}, \texttt{mi}, \texttt{in}\}$$

$$\texttt{metfromin} \rightarrow \{\texttt{me}, \texttt{et}, \texttt{tf}, \texttt{fr}, \texttt{ro}, \texttt{om}, \texttt{mi}, \texttt{in}\}$$

Five of eight pairs are shared; three differ. The transposition is now *visible to the substrate*, and graded recognition becomes possible: a near-miss resonates strongly but not identically, which is the behaviour §1.4 requires and the single-byte layer cannot deliver.

This changes nothing about the architecture's commitments. A bigram lane is one oscillator per ordered character pair, with displacement, velocity and natural frequency — the same `ByteLane` structure, a wider array. Specifically it introduces:

- no loss function, no gradient, no fitted parameter;

- no attention, no softmax, no learned projection;
- no probability semantics — lane energy is displacement in a field.

The distinction that governs this architecture is not *"are there numbers"* but *"where do the numbers come from."* A binding weight here is **computed** from a counted structural property of the lexicon by a stated formula; changing the lexicon changes it deterministically. A weight in a trained network is **fitted** by optimising a loss and admits no explanation beyond that optimisation. Descriptive measurement is not fitted inference — measuring a mass does not make mechanics machine learning.

n-gram indexing is also not novel-to-ML prior art: it originates in 1970s information retrieval (approximate string matching, spell correction) and predates statistical learning entirely.

4b.2 Measured result after implementing bigram lanes

Bigram lanes were implemented (`PhysicsEngine::bigram_bind_w`, `bind_bigram`). Grounding is identical in form to the single-byte version — inverse lane document-frequency across the lexicon — but over ordered pairs bracketed by BOS/EOS sentinels, so word-initial and word-final position are themselves observable pairs.

Recognition divides energy received by a concept by the geometric mean of its own binding mass and the query's:

$$\text{resonance}(c) \;=\; \frac{\sum_{g \in Q \cap B(c)} w_g}{\sqrt{M_c \cdot M_Q}}$$

This is energy per unit coupling — the same reason acceleration is force divided by mass — and it is why neither long nor short labels win by size. The earlier failures came from omitting exactly this normalisation.

Measured on the same 2,240-concept graph:

query	perturbation	recognised as	score	correct
`metformin`	none	`metformin`	1.000	✓
`metfromin`	transposition	`metformin`	0.686	✓
`insluin`	transposition	`insulin`	0.471	✓
`fissoin`	transposition	`fission`	0.621	✓
`fisssion`	insertion	`fission`	1.000	✓
`cavty`	deletion	`cavity`	0.830	✓
`tunel`	deletion	`tunnel`	0.885	✓

7 / 7, against 1 / 4 for single-byte lanes.

The `metformin` / `metfromin` pair is the decisive one: those two strings have **identical byte sets**, so the single-byte formulation could not distinguish them under any weighting. Over bigrams they share 5 of 8 pairs and differ in 3, and the correct concept is recovered at 0.686 against 1.000 for the exact spelling.

That gradation is the property §1.4 requires and a hash lookup cannot have: recognition degrades smoothly with distortion rather than failing discontinuously. It is also the evidence that recognition is not string matching — no comparison between the query and any stored label occurs at any point.

Status: bigram recognition is ENABLED in `codeq`, with an acceptance floor of 0.35 so unknown tokens resonate with nothing rather than matching noise. Single-byte recognition is retained in

the source only as the documented negative result. Per-query telemetry reports `lexical` versus `resonant` recognition counts, so residual dependence on exact lookup is a measured quantity rather than an assumption.

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5. Failure Modes

1. **Babble before consolidation.** With near-zero couplings the lane phases are unstructured and the sweep emits non-words. This is the correct initial state of a system that has not yet learned orthography, but it will read as garbage output until the lane↔lane offsets consolidate. Consolidation requires substantial exposure — Hebbian rather than gradient, but still genuinely a training process measured in thousands of turns, not tens.
2. **Phase resolution limits word length.** With K_{byte} sweep steps the angular resolution is $2\pi/K_{\text{byte}}$. A ten-character word requires ten sufficiently separated lane phases. Long words will be truncated or slurred until K_{byte} is raised, at proportional sweep cost.
3. **Homograph collision.** A single lane per byte value means the letter `a` has one phase at any instant. In a word requiring `a` at two different positions, the multi-pass sweep of §1.4 recovers the repetition only if the two occurrences are separated by other letters whose phases bracket them. Words like `aa` or vowel-adjacent repeats are expected to fail. A positional lane bank ($256 \times K$) would resolve this at a large state cost and would reintroduce something functionally close to positional encoding; it is deliberately not proposed here.
4. **No orthographic well-formedness guarantee.** Nothing in the mechanism prevents emission of byte sequences that are not words. Well-formedness is a statistical consequence of consolidation, not a constraint. This is a genuine difference from a grammar-based generator and should be reported as such rather than patched with a dictionary filter — a filter would reintroduce the vocabulary that MRS exists to remove.
5. **Reverse-gain instability.** Bidirectional coupling through a shared matrix creates a feedback loop: concepts drive lanes drive concepts. For η_{rev} above a critical value the loop has gain greater than unity and the field diverges. The VGHT §1.3 avalanche viscosity bounds this, but η_{rev} must be tuned conservatively and the loop gain monitored.

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6. Relation to the Existing Stack

MRS is the mirror of CPME one layer down. Every mechanism it uses already exists in the substrate for the concept layer:

Mechanism	Concept layer	Byte layer (MRS)
Chirality on input	`Cavity::t_arrival`	`ByteLane::t_arrival`
Momentum filter	`env_ema` on cavities	`env_ema` on lanes
Phase definition	$\text{atan2}(v_c, \omega_0 x_c)$	$\text{atan2}(v_b, \omega_0 x_b)$
Gauge offset	Ψ from `psi_for(rel)`	Ψ *learned* from observed phase
Phase-consistent Hebbian	CPME-II on tunnels	CPME-II on lane tunnels
Phase-collapse sweep	emits concept labels	emits **bytes**
Winner suppression	yes (concept appears once)	**no** (letters repeat)

The single genuinely new element is §1.1 reverse drive. Everything else is the same physics applied to a different oscillator population — which is the strongest argument that the architecture is

coherent rather than assembled.

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7. Summary

The CMD substrate was token-free on input and table-driven on output. MRS removes the remaining table by making the byte-lane layer bidirectional and equipping it with the chirality, consolidation, and phase-collapse mechanisms already built for concepts. Word forms become the projection of a byte-phase field rather than entries retrieved by graph key.

No tokeniser, no gradient, no layer stack, no next-symbol distribution. Storage obeys the Translation Capacity Lemma but is relocated from atomic surface strings into a factored coupling matrix, which is what makes morphological composition and novel forms reachable at all.

The honest cost is that consolidation is slow and early output is babble. A system that constructs its own orthography must first fail to.